Ch 2. Congruences (Disquisitions Arithmeticae by Gauss)

§2.1. Congruences

HW. 2, 6, 8, 14, 20, 27, 36, 45

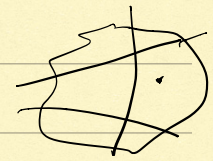
Def. Let $a, b, m \in \mathbb{Z}$ and $m \neq 0$.

$a \equiv b \mod m \iff m \mid (a-b)$. congruent 합동

$(\iff [a]_m = [b]_m)$

Thm ① $\equiv$ is an equivalence relation (반사, 대칭, 전이)

$(a \equiv a,\ a \equiv b \Rightarrow b \equiv a,\ a \equiv b \ \&\ b \equiv c \Rightarrow a \equiv c)$



② $+, -, \times$ 보존

$\left(\begin{array}{l} a \equiv_m b,\ c \equiv_m d \Rightarrow a \pm c \equiv_m b \pm d \\ \qquad\qquad\qquad\quad ac \equiv_m bd \end{array} \right)$ 法 m

$ac \equiv_m bc \Rightarrow a \equiv b$ if $(c, m) = 1$

pf) $m \mid bc - ac = c(b-a) \xrightarrow[\text{Euclid Lemma}]{(m,c)=1} m \mid (b-a) \therefore b \equiv_m a$.

Cor. Let $f(x) \in \mathbb{Z}[x]$. If $a \equiv b \mod m$, $f(a) \equiv f(b) \mod m$.

Def. Residues

① y is a residue of x modulo $m \iff y \equiv x \mod m$

② $\{x_1, \cdots, x_m\}$ is a complete residue system 잉여계

대표들의 집합 if $\forall$ integer y is congruent to one and only one $x_i \mod m$.

$\{0, 1, 2, \cdots, m-1\}$ $\{-\frac{m}{2}, \cdots, 0, 1, 2, \cdots, \frac{m}{2}-1\}$ m: even

$\{-\frac{m-1}{2}, \cdots, -1, 0, 1, \cdots, \frac{m-1}{2}\}$ m odd

③ $\{a - km \mid k \in \mathbb{Z}\}$: a residue class modulo m 잉여류

④ $\{x_1, \cdots, x_{\phi(m)}\}$: a reduced residue system modulo m

m과 서로소인 대표들 집합 if $x_i \not\equiv x_j$ (incongruent) $\forall i \neq j$ and $(x_i, m) = 1$

Euler totient fn $\rightarrow$ $\phi(m) :=$ # of positive integers $\leq m$ that are rel. prime to m

Thm (Fermat) p: prime $p \nmid a$. $a^{p-1} \equiv 1 \mod p$

Thm (Euler) $(a,m)=1$. $a^{\phi(m)} \equiv 1 \mod p$

pf) ① $\{r_1, \cdots, r_{\phi(m)}\}$ RRS $\Rightarrow \{ar_1, \cdots, ar_{\phi(m)}\}$ RRS

$(\because)$ $ar_i \equiv ar_j \overset{(a,m)=1}{\Rightarrow} r_i \equiv r_j \Rightarrow i=j$

$(ar_i, m) = (r_i, m) = 1$ ///

② $\prod_{i=1}^{\phi(m)} r_i \equiv_m \prod_{j=1}^{\phi(m)} ar_j = a^{\phi(m)} \times \prod_{j=1}^{\phi(m)} r_j \to m$과 서로소

$\Rightarrow 1 \equiv a^{\phi(m)} \mod m$ $\square$

Prop'n $(a,m)=1$. $\exists x \in \mathbb{Z}$ s.t. $ax \equiv 1 \mod m$. $\Rightarrow x = a^{-1} \mod m = [a^{-1}]_m$

Any two such x's are congruent modulo m.

If $(a,m)>1$, no such x exists.

pf) i) $(a,m)=1$. $\exists x, y \in \mathbb{Z}$ s.t. $ax+my=1 \Rightarrow ax \equiv 1 \mod m$

If $ax_1 \equiv ax_2 \equiv 1$, $ax_1 \equiv ax_2 \Rightarrow x_1 \equiv x_2$.

ii) Supp $d=(a,m)>1$. If $ax \equiv 1 \mod m$,

$\exists y \in \mathbb{Z}$ s.t $ax+my=1$. $d \mid a, m \Rightarrow d \mid (ax+my)=1$ $\therefore d=\pm 1$ $\square$ $*$

Prop'n p: prime. $x^2 \equiv 1 \mod p \Leftrightarrow x \equiv \pm 1 \mod p$

$(\because$ $p \mid (x^2-1) = (x-1)(x+1)$ $\overset{p \text{ prime}}{\Longrightarrow}$ $)$

proved by Lagrange

Thm (Wilson) p odd prime. $(p-1)! \equiv -1 \mod p$

pf) Any integer a with $1<a<p-1$ has $\bar{a}$ with $a\bar{a} \equiv 1 \mod p$, $1<\bar{a}<p-1$.

If $a \equiv \bar{a}$, $a\bar{a} \equiv a^2 \equiv 1 \Rightarrow a \equiv \pm 1$.

$(p-1)! = 1 \cdot (-1) \cdot (2\bar{2})(3\bar{3}) \cdots \equiv -1 \mod p$ $\square$

Cor p prime. $x^2 \equiv -1 \bmod p$ has solutions iff

$p = 2$ or $p \equiv 1 \bmod 4$. ($\Leftrightarrow 4 \mid \phi(p)$, $\phi(p) = p-1$)

pf) $p=2$ Trivial. Supp $p>2$ odd. $p \equiv 1$ or $3 \bmod 4$

If $p \equiv_4 3 \Leftrightarrow \frac{p-1}{2}$ odd, $x^2 \equiv -1$ implies $x^{p-1} = (x^2)^{\frac{p-1}{2}} \equiv -1$ *

If $p \equiv_4 1 \Leftrightarrow \frac{p-1}{2}$ even, $-1 \equiv (p-1)! = \prod_{j=1}^{(p-1)/2} j\,(p-j) \equiv \prod (-j^2)$

$1 \rightsquigarrow (-1)^{\frac{p-1}{2}} \left(\prod_{j=1}^{(p-1)/2} j\right)^2$

$\therefore \left(\prod_{j=1}^{(p-1)/2} j\right)^2 \equiv 1 \bmod p$

Lemma (Fermat) $p \equiv 1 \bmod 4$ prime.

$\exists\, a, b \in \mathbb{Z}$ s.t. $p = a^2 + b^2$